UNIT 10 · TOPIC 11

Environmental Implications of Precision Agriculture

Syllabus topic: Environmental implications of precision agriculture; spatial analysis and variable rate application

How PA reduces environmental impact

- Applying nutrients and pesticides only where, and in the amount, needed cuts over-application, nutrient RUN-OFF and LEACHING — protecting water bodies from eutrophication.
- Lower greenhouse-gas and energy intensity (Unit 7, Topics 11–12).
- Less soil compaction through controlled traffic (Unit 6, Topic 9).
- Water saved through variable irrigation.
- Section control and VRT curb chemical waste and drift (drones — Unit 2, Topic 12).

Net effect: higher input-use efficiency and profitability WITH a smaller environmental footprint — the core promise of PA.

Spatial analysis and VRA as the means

Interpolation, management zones and prescription maps (Topic 8) direct variable-rate application (Topic 9) so that each part of the field receives what it needs.

Limitations

- High initial cost.
- Data and skill requirements.
- Connectivity.
- Addressed by Custom Hiring Centres, drone-as-a-service and government schemes.

Practice MCQs

Q66. A key ENVIRONMENTAL benefit of precision agriculture is:

(a) More chemical use
(b) Reduced nutrient/chemical run-off and waste
(c) More tillage
(d) Higher emissions

Q67. Applying nutrients only where needed reduces:

(a) Yield
(b) Leaching and eutrophication of water bodies
(c) Soil fertility
(d) Rainfall

Q69. A major BARRIER to PA adoption by small farmers is:

(a) Too much rain
(b) High initial cost, data and skill needs
(c) Excess labour
(d) Low sunlight

Answers

Q66 — (b) Reduced nutrient/chemical run-off and waste — *PA lowers run-off, leaching, waste and emissions.*

Q67 — (b) Leaching and eutrophication of water bodies — *Less leaching → less eutrophication.*

Q69 — (b) High initial cost, data and skill needs

Sources: Srinivasan — Handbook of Precision Agriculture; ICAR/NIIT PA modules; Lillesand & Kiefer — Remote Sensing and Image Interpretation; ISO 11783 (ISOBUS); remote-sensing and NDVI references (as given in the source notes, July 2026); NPTEL precision-agriculture lectures.
